TOPIRAMATE

Teratogenic

Weight loss, kidney stones

CHEMICAL NAME 2,3:4,5-bis-O-(1-methylethylidene)-β-D-fructopyranose sulfamate
CLASS Sulfamate-substituted monosaccharide antiepileptic
TRADE NAMES Topamax · Topiramat · Trokendi XR · Qsymia (with phentermine)

THERAPEUTIC USES / INDICATIONS

- Focal and generalised tonic-clonic seizures; Lennox–Gastaut (adjunct).
- Migraine prophylaxis.
- Weight loss (combination product, some countries).

ADMINISTRATION

- **PO** with or without food; swallow tablets whole (bitter).
- Sprinkle capsules on soft food.
- Slow titration; maintain hydration.

DOSAGES

- **Epilepsy:** 25–50 mg at night for 1 week, increase by 25–50 mg weekly; usual 100–200 mg/day in 2 doses (max 500 mg).
- **Migraine:** 25 mg at night, increase to 50 mg twice daily.
- **Children:** 1–3 mg/kg/day, titrated to 5–9 mg/kg/day.
- **CrCl <70:** halve dose.

PHARMACOKINETICS

- **Peak:** 2 h. Bioavailability ~80%.
- Minimal metabolism; weak CYP2C19 inhibitor, weak 3A4 inducer.
- Excreted unchanged in urine. **t½:** 21 h.

NURSING MANAGEMENT

- Encourage 2–3 L fluid/day.
- Monitor bicarbonate, weight, mood, cognition.
- Report eye pain or sudden blurred vision urgently.
- Confirm contraception in women.

PHARMACODYNAMICS

- Blocks **sodium channels**, enhances **GABA-A**, antagonises **AMPA/kainate** glutamate receptors; inhibits carbonic anhydrase.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Paraesthesia, weight loss, anorexia.
- Cognitive slowing, word-finding difficulty, memory problems.
- Fatigue, dizziness, taste change.

ADVERSE EFFECTS (SERIOUS)

- Kidney stones; metabolic acidosis.
- Acute angle-closure glaucoma / myopia.
- Oligohidrosis and hyperthermia (children).
- Depression, suicidal ideation; teratogenicity (cleft lip/palate, low birth weight, neurodevelopmental).

CONTRAINDICATIONS & CAUTIONS

- Pregnancy (migraine prophylaxis) — and avoid in epilepsy unless no alternative.
- Women of childbearing potential not on highly effective contraception (migraine).
- Hypersensitivity.

KEY INTERACTIONS

- **Combined hormonal contraceptives:** ↓ efficacy at >200 mg/day.
- **Valproate:** hyperammonaemia.
- **Carbonic anhydrase inhibitors:** ↑ stones/acidosis.
- **Phenytoin, carbamazepine:** ↓ topiramate.

PATIENT EDUCATION

- Drink plenty of water.
- Report sudden eye pain/blurred vision, flank pain, mood changes.
- Tingling in hands and feet is common.
- Use highly effective contraception; plan pregnancy with doctor.

KEY POINTS

- Multi-mechanism antiepileptic; migraine prophylaxis.
- Kidney stones, glaucoma, acidosis.
- Paraesthesia, weight loss, cognitive slowing.
- Teratogenic.